Ravi Kasaudhan

Software Engineer

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PROFILE

AI Engineer with 4 years of experience building agentic AI, RAG, and real-time conversational voice and chat systems. Skilled in developing scalable cloud-native backends using Python, FastAPI, Kafka, Redis, Azure, and GCP, with expertise in LLM orchestration, tool calling, vector and hybrid search, and distributed architectures. Experienced in delivering reliable, low-latency AI solutions for automation, intelligent search, and customer engagement.

SKILLS

- **Programming Languages**: Python, Java, SQL
- **AI/LLM Frameworks & Libraries**: OpenAI, Google Gemini, Vertex AI, Pipecat, ADK, LangChain, Langgraph, Hugging Face, Scikit-learn, Pandas
- **Databases:** MongoDB, SQL Databases, Vector Databases
- **Cloud & Infrastructure:** Azure, Google Cloud Platform (GCP), Cloud Run, Kubernetes, Docker, Kafka, Google Pub/Sub, Redis, VPC Networking, Azure Functions/Logic Apps, Azure AI Search
- **Tools**: GitHub, Argo, Datadog, Power BI, Power Virtual Agents, CoPilot Studio, Yellow AI
- **Core Skills:** Generative AI, Agentic AI, RAG, Prompt Engineering, Function Calling, Vector Embeddings, Semantic & Hybrid Search, REST APIs, Async Programming, LLM Orchestration, LLM Evaluation, NLP, Machine Learning, CI/CD Pipelines, Microservices, Agile, MCP, A2A, Event-driven Architecture, Streaming Systems, System Design, Distributed Systems, Observability / Monitoring, Fault Tolerance

PROFESSIONAL EXPERIENCE

Nextiva, *Software Engineer* Aug 2025 – Present

- **Real-time AI Voicebot Runtime — *Python, asyncio, WebSockets, Deepgram, Gemini Live, Redis, GCP***
  - Built the core voicebot runtime that streams live call audio over WebSockets through a real-time VAD → STT → LLM → TTS pipeline, with barge-in interruption handling and turn detection.
  - Engineered a dual-mode design using composed STT → LLM → TTS and native speech-to-speech with Gemini Live, plus LLM tool-calling and a scalable process-per-call architecture on GCP with Redis coordination.
- **Agentic Conversational AI Editor — *FastAPI, WebSockets, Kafka, Redis, MongoDB, Anthropic/Claude***
  - Built an agentic AI editor that updates platform settings through natural language via LLM tool-calling, with specialist-based routing across AI Employees, skills, functions, workflows, handoffs, and knowledge sources.
  - Developed a local evaluation layer with scenario simulation, tool-call verification, and automated LLM-as-judge scoring to validate agent behavior.
- **Web Crawling & RAG Ingestion Pipeline — *Firecrawl, Gemini, Gemini-embedding-001, Redis***
  - Rebuilt the ingestion pipeline on Firecrawl, replacing a screenshot + HTML approach with Gemini and cutting ingestion cost by ~60% while improving throughput.
  - Generated 3072-dimensional embeddings with Gemini-embedding-001 and enabled hybrid-search retrieval with Redis hotloading, achieving <400 ms retrieval latency for knowledge queries.

EY, *AI Engineer* Sept 2024 – Aug 2025

- **Agentic IDP — *Semantic Kernel, GPT-4o, AI Builder, Dataverse, FAISS***
  - Developed an agentic document-processing solution to extract structured key-value data from unstructured and handwritten documents.
  - Implemented multi-model verification, Logprobs-based confidence scoring, dynamic prompt refinement, Dataverse mapping, and Ada/FAISS semantic search.
- **Video & PDF QnA — *Azure Video Indexer, GPT-4o, Azure AI Search, LangChain***
  - Developed a Video & PDF QnA system for querying video and document content.
  - Enabled real-time semantic search and reduced information retrieval time by 80%.

EY, *Associate AI Engineer* Aug 2023 – Aug 2024

- **Text2SQL — *GPT-3.5***
  - Developed a Text2SQL solution that generated SQL queries from database schemas and natural-language questions.
  - Built an automation flow to validate and execute queries and generate natural-language insights.
- **Translation & Summarization Bot — *Azure AI Translator, Azure OpenAI, Logic Apps, Functions, Yellow AI***
  - Built a multilingual translation and summarization bot using Azure AI Translator and Azure OpenAI GPT-3.5 Turbo.
  - Integrated Azure Logic Apps and Azure Functions with Yellow AI for deployment in Microsoft Teams.
- **AI Bots & RPA — *Microsoft Copilot Studio, Yellow AI***
  - Developed multiple AI-powered bots with integrated RPA solutions.
  - Deployed the bots in Microsoft Teams using Microsoft Copilot Studio and Yellow AI.

Anheuser-Busch InBev, *Intern* Jan 2023 – July 2023

- **Customer Engagement Chatbot — *Microsoft Bot Framework***
  - Designed a customer-facing chatbot using Microsoft Bot Framework.
  - Improved customer engagement by 40%.
- **GitHub Activity Dashboards — *Power BI***
  - Created Power BI dashboards to monitor GitHub repositories and activity.
  - Enhanced repository visibility and supported better decision-making.

PROJECTS

- **Excel & CSV / Product Search RAG System**
  FastAPI, MongoDB Atlas (Vector Search), Vertex AI Gemini 2.5 Flash, Gemini-embedding-001 (3072-dim), Pydantic, Asyncio , Hybrid Search
  - Developed a **RAG-based product search system** for Excel/CSV datasets using **MongoDB Atlas with hybrid retrieval** via **$vectorSearch** and **$search.**
  - Generated embeddings using **gemini-embedding-001 (3072-dim)** and used **Gemini-2.5-Flash** to convert natural language queries into validated JSON filters and aggregations
  - Built a **FastAPI retrieval service** with concurrent processing using **asyncio** and connection pooling to enable low-latency, scalable search.
- **Real Estate Price Prediction Project**
  NumPy, Pandas, Sklearn, Matplotlib
  - Designed and implemented a web-based real estate price prediction model leveraging the Bangalore home prices dataset (Kaggle).
  - Conducted comprehensive data cleaning and preprocessing using NumPy and Pandas, ensuring high-quality input data.
  - Developed a robust linear regression model with Sklearn, achieving accurate predictions on test data.

CERTIFICATIONS AND ACHIEVEMENTS

- Recipient of **COMPEX Scholarship by Indian Embassy** to study B. Tech CSE in VIT, Vellore Dec 2019
- Azure Data Fundamentals by Microsoft| link March 2024
- Power BI Data Analyst Associate | link Dec 2023
- Microsoft Certified: Azure AI Engineer Associate | link April 2025

EDUCATION

Vellore Institute of Technology Vellore, Tamil Nadu, India
B.Tech - Computer Science and Engineering *June 2019 – July 2023*

- CGPA : 8.46

ENGAGEMENTS AND INTERESTS

- Member of **Rotaract Club** of Central Lumbini, Nepal.
- Program Facilitator for “PAHUNCH” Project under Child Sensitive Social Protection (CSSP) at Community Family Welfare Association.